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Mitigating Class Imbalance in Pump-and-Dump Detection: A Comparative Analysis of Imbalance-Aware Algorithms

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ABSTRACT The cryptocurrency market has been subject to various forms of exploitation, with pump-and-dump schemes being a persistent problem that can lead to significant price volatility and financial losses. Previous studies have focused on predicting attack targets using machine learning approaches; however, these methods are often compromised by class imbalance and biases inherent in historical data. To address this limitation, we present an empirical framework for comparing manipulation-target prediction methods. We employ two-step bias-minimizing data normalization techniques to mitigate the effects of class imbalance and temporal heterogeneity. We compare established class-weighted, oversampling, and ranking approaches using a common event-level protocol. Validation Top@k%AUC selects CatBoost with Top@k%AUC early stopping (0.667). The reported model attains test Top@1 0.150, Top@10 0.563, and Top@k%AUC 0.674 on 80 events; tuned logistic regression has the highest descriptive AUC among the original candidates (0.697). We compare model AUC distributions using paired event bootstrapping and evaluate the validation-selected model in portfolio backtests incorporating transaction costs and price impact.

INDEX TERMS Classification, Imbalanced Datasets, Learning to rank, Machine Learning, Market Manipulation, Market Microstructure, Portfolio strategy

I. INTRODUCTION

A. WHAT ARE PUMPS AND DUMPS

PUMP-AND-DUMP schemes (P&Ds) are a type of fraudulent activity that involves artificially inflating the price of a cryptocurrency through coordinated buying efforts, followed by a rapid selling to unsuspecting investors. This phenomenon is often characterized by an abrupt and significant increase in trading volume, followed by a subsequent decline in price as the original buyers liquidate their positions.

These schemes typically involve experienced traders and market manipulators who orchestrate the operation through various channels, including online forums and social media platforms such as Telegram channels. The manipulation is designed to create a false sense of urgency and scarcity, which attracts inexperienced investors who may be unaware of the true nature of the event.

A typical pump-and-dump scenario can be observed in Fig. 1, which shows a clear example of this phenomenon. The figure illustrates price movements associated with a specific token VIB traded against BTC on Binance exchange, highlighting the abrupt spike in trading activity following the announcement of the token's release. This sudden increase in volume is often followed by a rapid decline in price as insiders sell their positions to new buyers.

B. RELATED WORK

The phenomenon of pump-and-dump schemes has been extensively studied in the context of cryptocurrencies, with research efforts spanning multiple approaches and methodologies.

One of the pioneering studies on this topic was [9], which introduced a comprehensive framework for understanding pump-and-dump operations. The authors highlighted the tar-



FIGURE 1. VIBBTC pump and dump manipulation carried out on Binance

geting of small market cap cryptocurrencies due to their lower trading volumes, utilizing data from 5 prominent crypto exchanges: Binance, Bittrex, Kraken, Kucoin, and Lbank. Their analysis covered pumps history of 1.5 years, with price and volume data being used as the primary sources. Subsequent research efforts have focused on leveraging machine learning (ML) algorithms to enhance P&D detection accuracy. La Morgia et al.'s work [10], [11] employed random forest algorithms, achieving improved results compared to traditional rule-based approaches. The authors in [4] further increased the effectiveness of La Morgia's work by incorporating Deep Neural Networks (LSTMs and Transformers) as anomaly detection algorithms. More recently, in [6] the authors built on these findings by studying a similar pump dataset between 2021 and 2022. They developed an innovative approach that employed resampling methods, such as SMOTE, combined with price and volume features and random forest classification to detect pump events 1 hour in advance.

In addition to pump detection, other research has focused on predicting the maximum price move during known pump events. Nghiem et al. [12] used market data and social sentiment with deep neural networks (LSTMs and CNNs) and achieved promising results. However, their analysis highlighted a potential limitation of using sentiment signals: historical bias may be introduced due to the lack of precise snapshots of historical social data.

A relevant example of work addressing the problem of predicting price manipulation targets is the study [16]. This research proposed an approach using random forest algorithms to identify potential pump and dump schemes among listed tokens. The authors analyzed a dataset of 180 pump events from the Cryptopia exchange, incorporating features such as price and volume data, global market capitalization data from Coinmarketcap, and token ratings by exchange. To enhance model robustness, the study employed filtering techniques to exclude pumps with suspicious announcement times that deviated significantly from full hours. Moreover, the authors validated their classification model through backtesting a portfolio strategy based on predicted pump targets. However, it is worth noting that the employed trading strategy relied on unrealistic assumptions, such as guaranteed profits of half the maximum price move during the pump and zero losses for false positive predictions.

A related study was conducted in [8], where authors enhanced their approach by incorporating channel embeddings into their model. They also utilized standard price and volume metrics, as well as external data sources such as market capitalization and social media information (e.g., Twitter followers, Reddit subscribers, Alexa rank). The analysis was performed on a dataset comprising 445 pump announcement events. However, due to the presence of duplicate pump events across multiple channels, the actual number of unique pump events studied is smaller than expected. Furthermore, the use of social data from different sources raises concerns about historical bias, as these metrics may have changed after the fact, potentially affecting the accuracy of the results.

Regarding market manipulation in a broad sense, several papers have also been published recently, for example [5], which studied the manipulation of trading volumes, also known as wash trading.

With the increasing popularity of ML applications in finance, in recent years we have seen many articles on predictions using the market microstructure, for example [2] and [13]. Promising results have also been published with applications of ranking models to finance [14]. Taking into account the rich accumulated experience in related areas, we aim to complement existing work with new factors based on the market microstructure.

In the broader ML literature, several families of techniques have been proposed for handling class imbalance beyond resampling, and each maps imperfectly onto the P&D target-prediction problem. Cost-sensitive learning [24] re-weights the loss so that misclassifying the minority class is more expensive; it does so without perturbing the feature space, which, as we show empirically in Section IV, can be an advantage when features are already cross-sectionally normalized. Focal loss [26], originally proposed for dense object detection, down-weights easy examples and concentrates the gradient on hard, informative ones; it has since been adopted for imbalanced tabular classification, but remains most effective when combined with deep models and large minority classes, conditions that do not apply to our 290 positive training events. Synthetic oversampling methods such as SMOTE [17] have known pathologies in high-dimensional settings [21], which is consistent with our results in Section IV.

Anomaly detection provides a complementary formulation that does not require balanced training data. Isolation Forest [27] and related one-class approaches have been applied to fraud and market-abuse detection when positive examples are essentially unavailable. In our setting, however, positive examples *are* available (at least in limited numbers) and the task is explicitly *cross-sectional*: we do not ask "is

this observation anomalous in isolation?" but rather "which asset in this cross-section is the most likely manipulation target?". This ranking-within-cross-section structure makes the problem closer to learning-to-rank than to unsupervised anomaly detection, and motivates the imbalance-aware ranking metric (Top@k%AUC) that we adopt as the training target throughout this paper. Our comparison combines cross-sectional panel normalization, microstructure features, and an imbalance-aware ranking objective for the one-positive-versus-many-negatives event structure.

This paper builds upon previous work in [16] and [8], which focused on predicting manipulation targets among traded assets. We identify several critical limitations in data preparation that limit model reliability and evaluate established remedies under a common event-level protocol. Specifically, we combine microstructure-based features using insights from areas of market prediction [2], [13] and wash trading detection [5]. Moreover, we explore the use of boosted trees, as implemented in CatBoost [15], which has been shown to outperform other algorithms on tabular data (e.g., [7]), and we compare their performance to the random forest model introduced in [3].

To further improve the classification models' performance, we also perform a cross-sectional normalization method for panel data. Moreover, we formulate a specialized metric tailored to optimize the classification model for highly unbalanced datasets, which was used for model training and validation.

The contribution is empirical integration and comparison, rather than a new learning algorithm: event-level ranking evaluation, common microstructure predictors, and explicit execution sensitivity on a documented Binance BTC-pair sample. Leakage prevention requires auditing preprocessing as well as chronological splits [29], [30]; recent imbalance-learning surveys likewise distinguish data-level resampling from algorithm-level weighting [31]. In cryptocurrency markets, quoted fees alone need not capture trading costs: changes in spreads and depth can offset fee reductions [32]. These distinctions motivate the separation of predictive evaluation and simulated execution below.

Table I summarizes the key methodological differences between our framework and the most closely related prior studies.

II. DATA COLLECTION AND PREPROCESSING

A. LABELED P&DS DATA

The released event list contains 498 unique announced P&D targets on Binance spot BTC pairs between January 3, 2018 and August 14, 2022. Each record supplies the target pair, exchange, and UTC announcement timestamp. These announcement labels are the fixed input to the empirical pipeline; no event is retained or discarded according to its subsequent return.

a: Channel identification.

We identified eight candidate Telegram channels through a two-step procedure. First, we searched Telegram directly using the keywords "crypto pump," "pump signal," "pump and dump," and their localized equivalents, retaining public channels whose stated purpose was the coordination of pump events. Second, we cross-referenced the resulting list with channels named or cited in prior literature [6], [10], [16]. For each channel, we exported the full message history (JSON format) using the Telegram Desktop client's built-in export functionality. The channel names cannot always be redistributed, but the timestamps and tickers of the retained events are in the public event file.

b: Labeling procedure.

The operational label record contains one row per {ticker, exchange, announcement timestamp}; there are no exact or ticker–timestamp duplicates. The repository does not contain Telegram message text or message URLs, so the released event labels support exact computational reproduction of the experiments but not an independent reconstruction of the original social-media annotation decisions.

c: Inclusion criteria.

A manifest event enters a model sample only if *all* of the following data-availability conditions hold:

- the target is a Binance spot pair quoted in BTC;
- the target has raw trade data in the exact past-only eligibility interval $[T - 15\text{min} - 1\text{d}, T - 15\text{min})$;
- a feature cross-section can be materialized from the local Binance archive.

d: Exclusion criteria.

No post-announcement price or volume threshold is used for sample inclusion. Events without the raw target history required to construct a feature cross-section are excluded and recorded separately. These records document data availability, not social-media rejection reasons.

We performed the following preprocessing steps:

1) parse and validate every manifest record;
2) discover candidate BTC pairs only from date partitions intersecting the past eligibility interval;
3) inspect actual trade timestamps rather than date-partition membership, and require at least one trade in $[T - 15\text{min} - 1\text{d}, T - 15\text{min})$ for each candidate and the announced target;
4) materialize features and split complete event cross-sections chronologically.

The resulting dataset contains 473 unique market manipulation events. Restricting the analysis to Binance BTC pairs is a deliberate design choice with known trade-offs: (i) we lose events that targeted USDT-quoted pairs, tokens listed only on smaller venues (e.g., KuCoin, Gate, LBank, Cryptopia before its 2019 shutdown), and decentralized-exchange pumps; (ii) in exchange, we obtain full trade-level history for

TABLE I. Comparison of the proposed framework with closely related prior work on P&D target prediction

Study	Exchange	# Events	Model	Imbalance Handling	Backtesting
Xu & Livshits [16]	Cryptopia	180	Random forest	None	Unrealistic assumptions
Hu et al. [8]	Multiple	445	Seq. model + channel embed.	None	None
Fantazzini & Xiao [6]	Binance	Varies	Random forest	SMOTE	None
La Morgia et al. [10]	Multiple	1111	Random forest	None	None
Ours	Binance	473	CatBoost, LR, RF, Ranker	Cross-sectional norm., Top@k%AUC opt., class weighting	Impact-aware execution

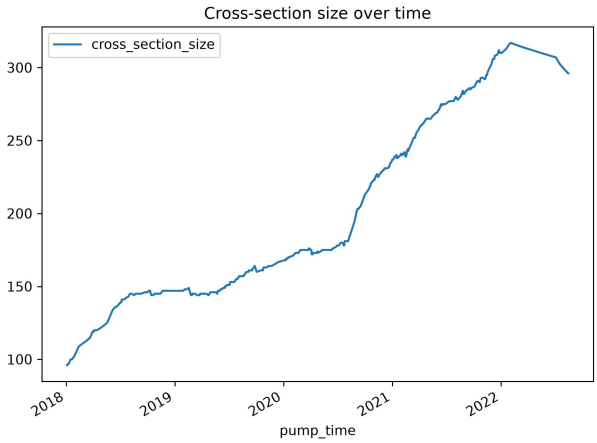


FIGURE 2. Binance has been increasing the number of tickers traded against BTC by listing more tokens.

every pair in the cross-section, which is the minimum requirement for computing slippage, flow-imbalance and power-law features consistently across the whole sample; (iii) the restriction is aligned with the most comparable prior work on Binance P&D target prediction [6] and on microstructure-based detection [10], so our results remain directly comparable. We further discuss the limits of external validity in Section VI. Our data presents a substantial challenge, with P&D events occurring relatively infrequently compared to normal trading activity. During our study period, the average eligible cross-section contains 195.05 instruments quoted in BTC. The historical dynamics of the number of listings is shown in Fig. 2.

We split the data into training, validation and testing samples by 2020-09-01 and 2021-05-01 respectively. Table II describes the resulting datasets after the split:

B. MARKET DATA

To enable comparison with existing research, we selected Binance as the sole exchange platform. We use Binance's public historical trade archive for every BTC-quoted pair

	Train	Validation	Test
Positive	290	103	80
Negative	44578	24274	22935
Total	44868	24377	23015
Avg. Cross-section Size	154.72	236.67	287.69

TABLE II. P&D Train-Validation-Test samples statistics

needed by the 2018–2022 event manifest and its 30-day feature lookbacks.

A representative example of our collected data is presented in Table III.

C. FEATURE ENGINEERING

We collected trade-level data as compressed daily archives for the required Binance pairs and dates. Because trading activity varies substantially across tickers and over time, the raw event streams are irregular and cannot be used directly as fixed-size inputs to standard machine-learning models. We therefore implemented a feature-generation pipeline that (i) decompresses and loads the relevant archives, (ii) partitions data into time-aligned windows, (iii) computes microstructure features summarizing pre-announcement order flow, and (iv) aggregates ticker-level features into an event-level cross-section associated with each pump. The pipeline is:

1) Download trade-level data from Binance's public daily archive.
2) For each pump event, construct the cross-section from tickers with at least one trade in the exact interval $[T - 15\text{min} - 1\text{d}, T - 15\text{min})$. Partition membership alone is insufficient because a daily partition can contain trades later than the decision time.
3) For each eligible ticker, load 30 days of history ending at the decision cutoff and compute features over predefined half-open lookback windows $[T - 15\text{min} - x, T - 15\text{min})$. Hourly normalizer bars are re-anchored to the cutoff so every included bar is strictly pre-decision.
4) Combine features for all tickers into a single event-level

	price	qty	time	isBuyerMaker	quote
0	0.000002	2437.00	2022-07-15 15:00:02.422000	True	0.004240
1	0.000002	275.00	2022-07-15 15:00:02.434000	False	0.000479
2	0.000002	1125.00	2022-07-15 15:00:22.884000	True	0.001946
3	0.000002	941.00	2022-07-15 15:00:52.277000	True	0.001628
4	0.000002	184.00	2022-07-15 15:00:52.277000	True	0.000318

TABLE III. Structure of collected data: *price* shows the price that the order was filled at, *qty* is the amount of base asset traded, *isBuyerMaker* indicates if the buyer's order related to this transaction was already in the orderbook: if True, the trade is classified as a Sell event and if False the trade is characterized as a Buy event.

dataset and assign labels by adding *is_pumped*, which equals 1 for the pumped ticker and 0 otherwise.

This modular pipeline supports reproducible feature construction with explicit decision-time boundaries. All models use the same frozen feature inputs, with a feature file available for each of the 473 retained events. The implementation and reproduction instructions are available in [18], and the reproduction bundle records the exact inputs, tests and outputs described below.

1) Imbalance features

Volume imbalance measures which side is more prevalent in trade data. It is the ratio of signed notional in the quote asset (negative for seller-initiated trades and positive for buyer-initiated trades) to total absolute quote notional. Values near 1 indicate predominantly buyer-initiated flow and values near -1 predominantly seller-initiated flow.

$$\text{Volume Imbalance} = \frac{\sum_{t\in\mathcal{T}} s_t p_t q_t}{\sum_{t\in\mathcal{T}} p_t q_t}, \tag{1}$$

where $p_t q_t$ is quote notional and $s_t \in \{-1, 1\}$ is the taker-side sign at time t.

2) Quote slippages

We also created features based on slippage. Since we do not have information about order types, we cannot determine whether a trade was executed via a market or limit order; therefore, we treat slippage as a useful proxy for measuring market participants' impatience. We follow the methodology used in prior work by grouping ticks by execution time. Given that we deal with highly illiquid markets—where the time between trades can be as long as 15 minutes—we can reasonably assume that aggregated ticks correspond to a single order. This allows us to estimate the price impact of the aggregated order. The intuition is as follows: if a market participant is willing to incur large slippage losses by submitting a large buy order, they likely want to build their position quickly, potentially in anticipation of the pump. Based on this idea, we compute slippage for each aggregated trade using the formula in (2) below:

$$\begin{aligned} L &= \left| \sum_{i=1}^{N} q_i P_i - P_0 \sum_{i=1}^{N} q_i \right|, \\ L_{\text{signed}} &= L \ \text{sign}\left(\sum_{i=1}^{N} s_i q_i \right). \end{aligned} \tag{2}$$

where q_i, P_i, and s_i are respectively the unsigned base quantity, execution price, and taker-side sign of the ith print in an aggregated timestamp group, and P_0 is its first execution price. Thus L is the absolute difference between observed quote notional and the notional implied by filling the whole group at its first price; the sign is attached from net buyer- or seller-initiated base quantity. The slippage-imbalance feature is $\sum L_{\text{signed}} / \sum L$. As an order consumes liquidity it can appear as a sequence of prints at worsening prices, so this statistic is intended to proxy how aggressively the group walks available liquidity.

3) Arithmetic return features

For each horizon x, the implementation computes the arithmetic return $10^4(p_{\text{last}}/p_{\text{prev}} - 1)$ over $[T - 15\text{min} - x, T - 15\text{min})$. Return z-scores compare that window with the mean and sample standard deviation of decision-anchored hourly returns over the preceding 30 days. For $x \in \{5, 15\}$ minutes, the exact-window return is converted to an hourly rate before standardization; for horizons of at least one hour, complete cutoff-anchored hourly bars are averaged. Thus the 5-minute, 15-minute, and 1-hour columns are distinct, while every numerator and normalizer remains strictly pre-decision.

4) Power law features. Quantile spread of order sizes

To summarize the tail of aggregated trade notionals $x \geq x_{\min}$ within each lookback window, we use the Pareto density

$$f(x; \alpha, x_{\min}) = (\alpha - 1) x_{\min}^{\alpha-1} x^{-\alpha}, \qquad \alpha > 1, \tag{3}$$

where $x_{\min}$ is the minimum positive aggregated notional observed in the window. The code uses the closed-form maximum-likelihood estimate

$$\hat{\alpha} = 1 + \frac{n}{\sum_{i=1}^{n} \log(x_i / x_{\min})}. \tag{4}$$

Smaller $\hat{\alpha}$ indicates a heavier notional tail. Before cross-sectional standardization, the implementation clips $\hat{\alpha}$ to the

closed interval $[1, 2]$. This deterministic bound handles degenerate short windows in which the denominator is zero (and the raw estimate is infinite) and limits the influence of extremely sparse tails. This is a fixed model-preprocessing rule, not a sample-dependent 95% winsorization; no histogram-density regression or log–log OLS is performed.

All of the features discussed are further explained below (see Table IV):

D. CROSS-SECTIONAL APPROACH

Given that our event dataset spans early 2018 through August 2022, we account for temporal heterogeneity by applying cross-sectional standardization. This approach reduces market-regime level shifts and focuses the models on relative cross-asset dynamics.

We apply cross-sectional standardization to the unbounded return, return-z-score, quote-volume-z-score, and power-law features. For each selected feature, we subtract its mean within an event and divide by its within-event sample standard deviation:

$$X_{\text{standardized}} = \frac{X - \bar{X}_{CS}}{\bar{\sigma}_{CS}} \tag{5}$$

If a selected feature is constant within an event, it contains no relative ranking information and is mapped to zero; all-missing event columns remain missing for native handling or train-only imputation. We do not apply this second-stage transform to bounded imbalance/share features or to count features (trade counts and lifetime prior-pump count), whose nonnegative levels retain their direct interpretation. Fig. 3 demonstrates the transform on a sample of five cross-sections for three selected numerical features:

III. MODELLING

In this section, we attempt to create a model able to label this fraudulent activity 15 minutes before the announcement. As discussed earlier this is a challenging task due to high imbalance in the data. The task of finding manipulated assets can be designed as a binary classification problem where "1" implies that the asset is pumped and "0" that it is not. Therefore, we trained logistic regression, random forest, and CatBoost classifiers. The problem can also be formulated as learning to rank: the CatBoost ranker is supervised by each asset's realized arithmetic return over the first five minutes after announcement, ranked highest-first within its event cross-section (ordinal position 1 is the highest realized return). These post-announcement returns are labels only; the ranker's regressors are the same strictly pre-decision microstructure features used by the classifiers.

A. EVALUATION METRICS

Top@k is the probability that the ground-truth target occurs among a model's k highest scores. In portfolio terms, it is the probability that a size-k portfolio contains the pump target. Top@k% replaces the fixed count with a percentage of each cross-section, controlling for the growth in Binance listings described in Section II-A. We use one notation convention throughout: Top@k and Top@k% denote the fixed-count and percentage hit-rate curves; Top@kAUC and Top@k%AUC denote their respective areas. A concrete threshold replaces k in a point label, so Top@10 is the top-10 hit rate and Top@5% is the top-5% hit rate. The reported and optimized area in this study is Top@k%AUC; Top@kAUC is reserved for an area over fixed-count k and is not reported here.

The area under the Top@k% curve (Top@k%AUC) serves as our primary optimization target. In our setup we integrate the Top@k% curve over the low-$k\%$ region $k\% \in (0,\ 20\%]$ and normalize by the length of the integration range so the metric remains in $(0, 1)$:

$$\text{Top@k\%AUC} = \frac{1}{0.20}\int_0^{0.20} \text{Top@k\%}\, d(k\%). \tag{6}$$

Restricting integration to the $(0, 20\%]$ region is motivated by two considerations. First, it is the economically relevant region for a concentrated P&D portfolio: a practical strategy holds at most the top 10–20% of the cross-section in any given event. Second, all reasonably-trained models saturate above $k\% \approx 30\%$ and their Top@k% curves converge, so integrating over the full $(0, 1)$ range dilutes the metric with a large k region where every model is close to 1 and no longer discriminates between them. Truncating at 20% makes Top@k%AUC materially more sensitive to differences between models and to hyperparameter choices, which is the property we want from both an Optuna tuning target and an early-stopping signal inside CatBoost. Numerically, Top@k%AUC values reported in this paper are therefore not directly comparable to AUCs integrated over $(0, 1)$; a model that saturates quickly below 20% can attain a Top@k%AUC close to 0.8 under this convention.

1) Limitations of Top@k-based metrics

While Top@k metrics are directly relevant for portfolio construction (how many assets must we hold to catch the pump?), they have limitations. They do not measure overall classification quality across the full score distribution and can mask poor calibration. A model may achieve high Top@k by ranking the pump target above most non-targets without producing well-separated probabilities. To address this, we complement Top@k with standard classification metrics.

2) Standard classification metrics

We additionally report three standard metrics on both the validation and test sets:

- **PR-AUC:** Area under the precision–recall curve, which is more informative than receiver operating characteristic (ROC) AUC under extreme class imbalance [24].
- **F1-score:** Harmonic mean of precision and recall, computed using a "Top@1 per cross-section" decision rule (predicting the highest-scored asset as positive).

Feature	Description	Notation
Asset return	return of the asset calculated over the window of size (T-15min-x, T-15min) prior to the pump announcement	*asset_return*[x]@t
Standardized return	exact-window hourly-rate return (5m, 15m) or mean cutoff-anchored hourly return ($x \geq$ 1h), minus the 30-day hourly mean and divided by its standard deviation	*asset_return_zscore*[x]@t
Standardized volume in BTC	exact-window hourly-rate volume (5m, 15m) or mean cutoff-anchored hourly volume ($x \geq$ 1h), standardized by 30-day hourly moments	*quote_abs_zscore*[x]@h
Share of long trades	share of long trades as the ratio of number of long trades and all trades of this time window (T-15min-x, T-15min)	*share_of_long_trades*[x]@h
Number of trades	number of timestamp-aggregated trades within $[T - 15\text{min} - x, T - 15\text{min})$	*num_trades*[x]@h
Flow imbalance ratio	volume imbalance ratio computed over the window of size (T-15min-x, T-15min)	*flow_imbalance*[x]@h
Quote slippage imbalance ratio	imbalance ratio of slippages in BTC calculated over the same time window	*slippage_imbalance*[x]@t
Powerlaw alpha	estimated alpha from powerlaw probability density function using the window of size (T-15min-x, T-15min) prior to the pump	*powerlaw_alpha*[x]@h
Number of previous pumps	lifetime number of manifest events for this token strictly before the current announcement	*num_prev_pump*

TABLE IV. These are the features included in all models. These features are used to describe historical dynamics of price and trading volume in the run up to the pump. $x \in \{5m, 15m, 1h, 2h, 4h, 12h, 1d, 2d, 7d, 14d\}$

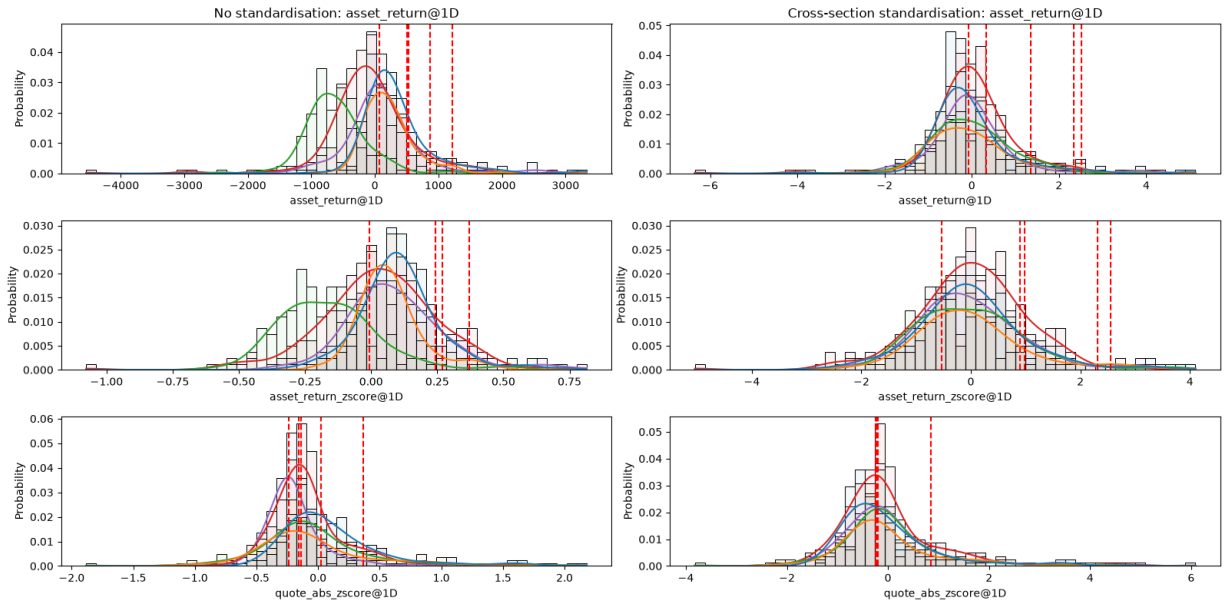


FIGURE 3. Features with and without cross-sectional standardization: histograms and kernel-density estimates for five P&D events. Applying standardization helps to align distributions more closely with one another across pumps.

- **Balanced accuracy:** Average of sensitivity and specificity, providing a class-imbalance-aware accuracy measure.

B. BASELINE MODEL

We use logistic regression as a simple baseline and compare it with a random-ranking model that selects candidate targets uniformly within each event cross-section. The tuned L1 baseline uses the `liblinear` solver and retains the saved validation-tuned $C = 0.001880$ and positive-class weight 72.550 (negative-class weight 1). We additionally compare L2-penalized logistic regression with the same predictors and class weights. All logistic models are fitted on training rows only. Our primary metrics are Top@k and Top@k%AUC,

discussed in Section IV.

C. TRAINING

We used the training set to train the models. To reduce overfitting, tree depth was limited through the *max_depth* hyperparameter for both random forest and CatBoost models. We also applied early stopping for both CatBoost classifier and CatBoost ranker and used validation set to compute evaluation metrics. We created a custom evaluation metric Top@k%AUC, this metric was used by the trainer to find the optimal number of boosted trees. We also set the number of early stopping rounds to 50 which implies that there is no improvement in Top@k%AUC for more than 50 rounds, the training process will stop and the model will rollback to the best state. This was used in the training process of the model CatBoost classifier + Top@k%AUC early stopping (early stopping model with custom evaluation metric).

Hyperparameters were originally optimized using 100 completed Optuna trials per pipeline on the historical validation period [1]. For the additional analyses we retain those saved parameter sets and refit the historical tuned candidates on the training sample without a new search. An additional L2 logistic comparator selects only C from a fixed 25-value logarithmic grid from 10^{-6} to 10^{2} on validation Top@k%AUC, with the L1 class weights held fixed and ties resolved toward smaller C. It uses the same preprocessing and `liblinear` solver, with tolerance 10^{-8} and at most 2000 iterations; a convergence warning stops the search. The selected L2 value is $C = 4.642 \times 10^{-5}$. This additional comparison follows earlier test inspection and remains exploratory. The original validation-AUC rule favors CatBoost + ES; validation is also used for early stopping, so its score is selection evidence rather than an unbiased performance estimate. We select the candidate with the highest validation Top@k%AUC across all models and use that fitted model for portfolio analysis. This rule selects CatBoost + ES; descriptive test rankings do not change that decision. The test period has been inspected in earlier research, so the resulting comparisons remain exploratory.

We also used SMOTE proposed by [17] to tackle the problem of class imbalance. It allows to create new observations for the minority class to balance the dataset. Instead of copying existing minority points, it generates new, slightly different ones by interpolating between neighbors.

IV. RESULTS

A. VALIDATION PERFORMANCE AND MODEL SELECTION

Table V reports the tuned candidates from the original validation comparison. CatBoost + ES has the highest validation Top@k%AUC (0.667), ahead of tuned CatBoost (0.655) and tuned random forest (0.628). Its validation Top@1, Top@5, and Top@10 are 0.175, 0.427, and 0.583; Top@1%, Top@5%, and Top@10% are 0.320, 0.641, and 0.718. It shares the highest F1 (0.175) and balanced accuracy (0.586), and leads PR-AUC (0.098). CatBoost + ES is the portfolio model because it maximizes the validation criterion; the bootstrap does not establish superiority over logistic regression on new data.[1]

Fig. 4 shows validation Top@k against the validation-specific random-ranking baseline, while Fig. 5 shows validation Top@k% curves and their Top@k%AUC values. Fig. 6 provides the corresponding validation classification comparison. Together with Table V, these figures make the selection decision auditable without consulting the test set.

B. TOP@K ACCURACY

We report test Top@k accuracy for the tuned candidates. The validation-selected CatBoost + ES model is retained for portfolio analysis; descriptive test comparisons do not change the selection rule.

Table VI shows that logistic regression remains a competitive baseline, indicating that the engineered features contain predictive signal. CatBoost + ES leads at $k = 1$ (0.150); tuned CatBoost leads at $k = 2$ (0.275); and tuned logistic regression leads at $k = 5$ (0.425), $k = 10$ (0.575), $k = 20$ (0.700), and $k = 30$ (0.763). The tuned SMOTE variant remains below the leading class-weighted models across the reported thresholds, suggesting that synthetic oversampling is poorly aligned with the cross-sectional microstructure features used in this setting. Fig. 7 reports the corresponding Top@k curves.

All trained models outperform the random-ranking baseline, indicating that the feature set contains predictive information about future manipulation targets. Top@k values for the random-ranking model are computed as follows:

$$\text{Top@k}_{\text{random}} = \frac{1}{N}\sum_{i=1}^{N}\frac{k}{T_i} \tag{7}$$

where N is the number of cross-sections in the test set and T_i is the i-th cross-section size. This is evidence of predictive signal within this sample. With its saved tuned parameters, the ranker remains below the leading classification models on the aggregate ranking metric (test Top@k%AUC 0.629); the mechanism is analyzed later in this section.

C. TOP@K% ACCURACY

Table VII reports the corresponding Top@k% pattern. Tuned CatBoost leads at $k\% = 1\%$ (0.338). Tuned logistic regression leads at $k\% = 2\%$ (0.475), 5% (0.638), 10% (0.750), 20% (0.863), and 50% (0.963). The aggregate ranking differences are assessed in Section IV-I.

D. TOP@K%AUC

In order to control for the problem of various cross-section sizes, we want to compute Top@k% instead of regular Top@k. Taking the normalized area under the Top@k% curve over $k\% \in (0, 20\%]$ allows us to compare models in the economically relevant low-portfolio-size region, consistent

[1]The original model-comparison and CatBoost robustness estimates are retained as reported. The seven-model bootstrap and random-portfolio comparison use refitted models and their saved predictions; their intervals describe those fits.

TABLE V. Validation metrics and final model selection. Top@k%AUC is the selection criterion; the maximum in each metric column is shown in bold. F1 and balanced accuracy use the Top@1-per-cross-section decision rule.

	Top@k%AUC	Top@1	Top@5	Top@10	Top@1%	Top@5%	Top@10%	PR-AUC	F1	Balanced accuracy
CatBoost classifier + Top@k%AUC early stopping	**0.667**	**0.175**	**0.427**	**0.583**	0.320	**0.641**	**0.718**	**0.098**	**0.175**	**0.586**
CatBoost classifier + Tuned	0.655	0.165	0.417	0.524	0.301	0.592	0.709	0.087	0.165	0.581
Random forest + Tuned	0.628	**0.175**	0.398	0.485	**0.330**	0.534	0.689	0.085	**0.175**	**0.586**
Logistic regression + Tuned	0.616	0.107	0.301	0.485	0.223	0.563	0.680	0.059	0.107	0.552
CatBoost ranker + Tuned	0.550	0.107	0.301	0.466	0.223	0.476	0.602	0.048	0.107	0.552
CatBoost classifier + SMOTE + Tuned	0.524	0.107	0.320	0.437	0.233	0.456	0.544	0.060	0.107	0.552

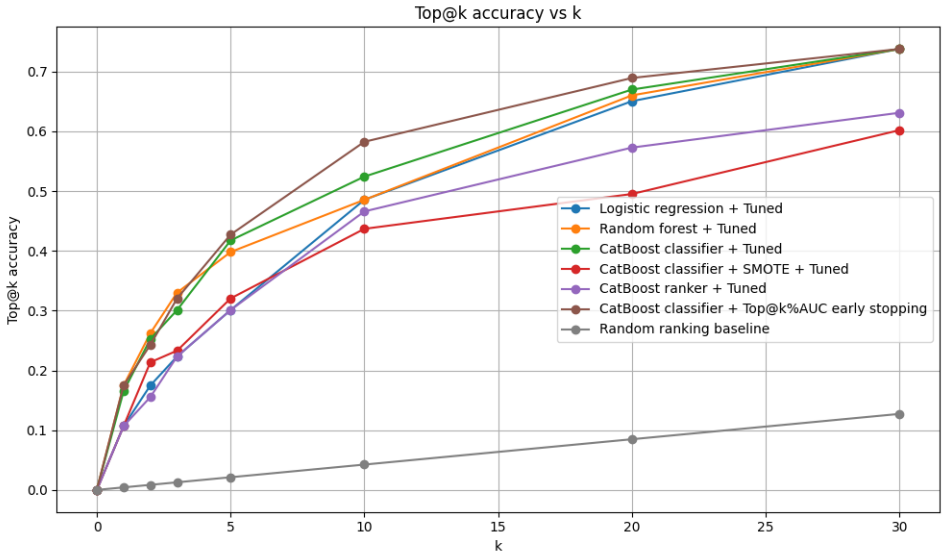


FIGURE 4. Top@k accuracy on the validation sample for each tuned baseline, CatBoost + ES, and the validation random-ranking baseline. Validation supplies the original selection criterion.

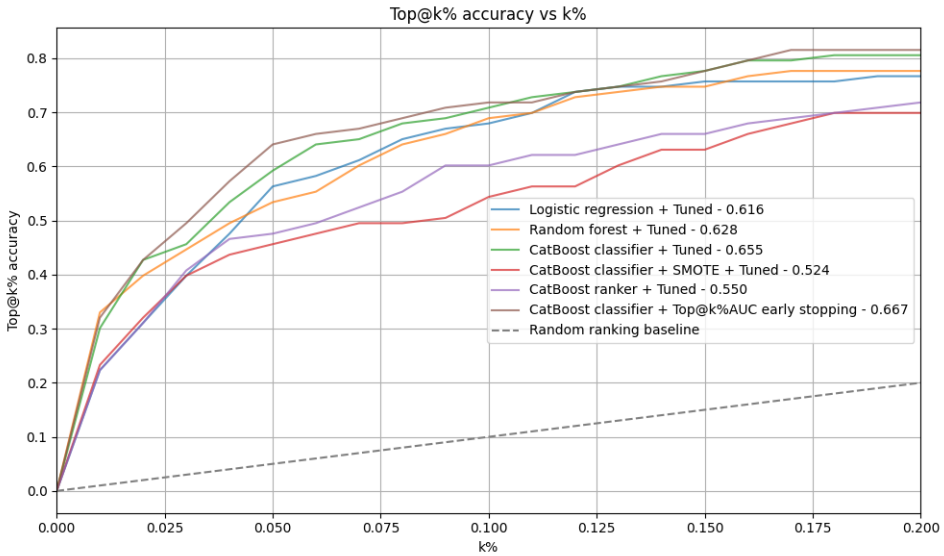


FIGURE 5. Top@k% accuracy and Top@k%AUC on the validation sample. CatBoost + ES has the largest validation Top@k%AUC (0.667) and selects CatBoost + ES for portfolio analysis.

with Eq. 6. In Fig. 8 we compare the models based on this metric.

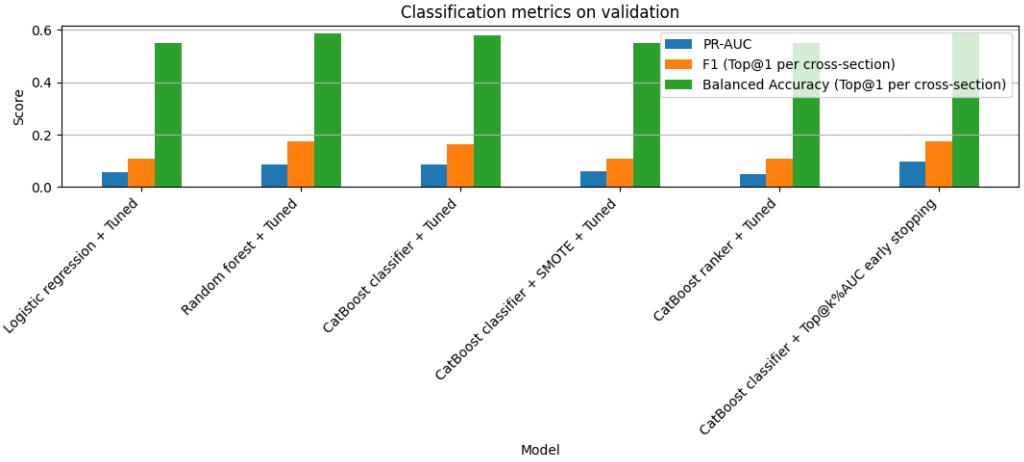


FIGURE 6. PR-AUC, F1, and balanced accuracy on the validation sample. F1 and balanced accuracy use the Top@1-per-cross-section decision rule.

TABLE VI. Top@k accuracy of models on test sample

Model	Top@1	Top@2	Top@5	Top@10	Top@20	Top@30
Logistic regression + Tuned	0.100	0.200	**0.425**	**0.575**	**0.700**	**0.763**
Random forest + Tuned	0.113	0.175	0.350	0.500	0.650	0.713
CatBoost classifier + Tuned	0.138	**0.275**	0.400	0.525	0.650	0.725
CatBoost classifier + SMOTE + Tuned	0.113	0.163	0.300	0.425	0.538	0.600
CatBoost ranker + Tuned	0.063	0.163	0.275	0.463	0.613	0.700
CatBoost classifier + Top@k%AUC early stopping	**0.150**	0.225	0.388	0.563	0.663	0.750

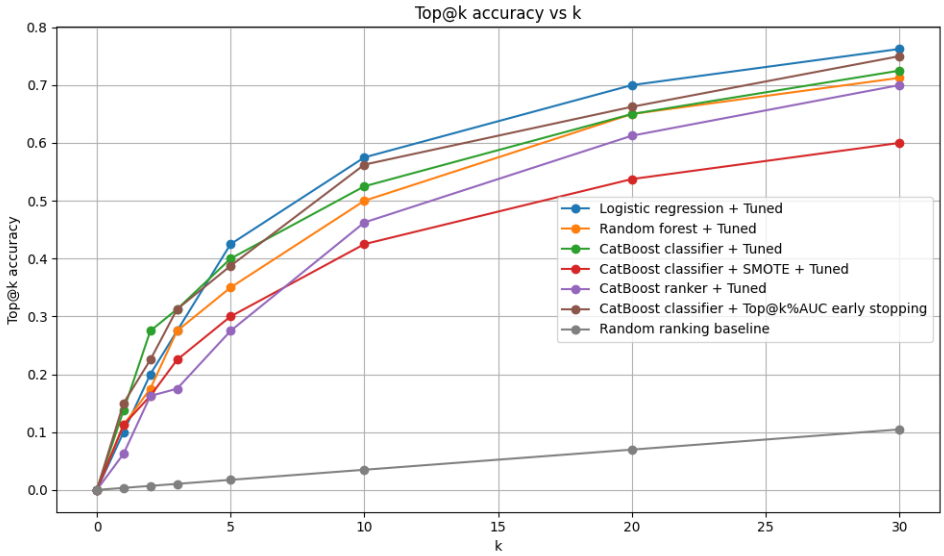


FIGURE 7. Top@k accuracy on the test sample for each tuned baseline, CatBoost + ES, and the random-ranking baseline. The horizontal axis is the portfolio size k; the vertical axis is the empirical probability that the manipulation target appears among the k highest-scored assets. Table VI reports the tuned variants.

E. CLASSIFICATION METRICS

We complement the ranking-based Top@k metrics with three standard classification metrics (defined in Section III-A): PR-AUC, F1-score (Top@1-per-cross-section decision rule), and balanced accuracy. PR-AUC is preferred over ROC-AUC at this imbalance ratio [24]. Fig. 9 reports all three for the tuned baselines and CatBoost + ES. Among these models, tuned CatBoost leads PR-AUC (0.0866), followed by CatBoost + ES (0.0836). CatBoost + ES leads F1 (0.1500) and balanced accuracy (0.5735), followed by tuned CatBoost at 0.1375

TABLE VII. Top@k% accuracy of models on test sample

Model	Top@1%	Top@2%	Top@5%	Top@10%	Top@20%	Top@50%
Logistic regression + Tuned	0.325	**0.475**	**0.638**	**0.750**	**0.863**	**0.963**
Random forest + Tuned	0.275	0.425	0.575	0.700	0.825	0.925
CatBoost classifier + Tuned	**0.338**	0.425	0.625	0.725	0.813	0.950
CatBoost classifier + SMOTE + Tuned	0.238	0.325	0.513	0.588	0.763	0.900
CatBoost ranker + Tuned	0.200	0.375	0.525	0.700	0.825	0.888
CatBoost classifier + Top@k%AUC early stopping	0.313	0.463	0.600	0.738	0.838	0.950

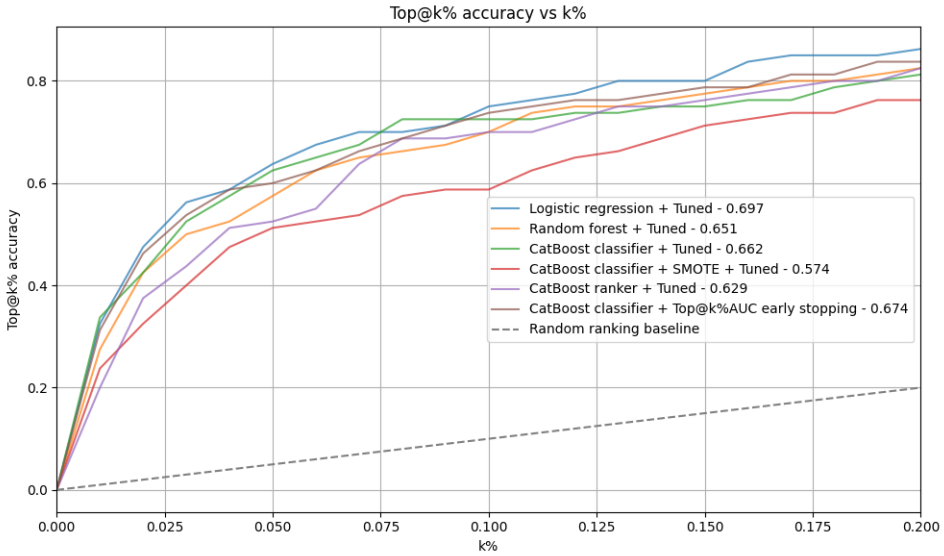


FIGURE 8. Top@k% accuracy on the test sample, normalizing the portfolio size by the cross-section size to control for its growth over time. The horizontal axis is the fraction *k*% of the cross-section retained; the vertical axis is the probability that the target is covered.

and 0.5672. This pattern is consistent with the strong low-k ranking performance of both CatBoost variants.

F. ANALYSIS OF SMOTE PERFORMANCE

Tuned CatBoost with SMOTE performs worse than the leading class-weighted classifiers, attaining Top@1 of 11.25% and Top@k%AUC of 0.574; tuned logistic regression reaches 0.697. Three properties of the panel offer plausible explanations, rather than experimentally isolated causes, for the remaining deficit.

1) **Bounded and sign-meaningful features.** Several of our features—flow imbalance, slippage imbalance, share of buy-initiated trades—are naturally bounded in $[-1, 1]$ (or $[0, 1]$) and are informative primarily near the positive extreme (e.g., heavy buy-side imbalance just before a pump). Interpolating between two pumped samples whose imbalance is $+0.85$ and another whose imbalance is -0.20 yields a synthetic "pump" with imbalance $+0.32$, which lies inside the bulk of the negative-class distribution. SMOTE therefore dilutes rather than reinforces the class-discriminative region of the feature space.
2) **Cross-sectional reference.** The unbounded return, standardized-volume, and tail-shape features are standardized within the event cross-section (Eq. 5); for those columns, a value of $+2$ means "two cross-sectional standard deviations above the same-event average." Bounded ratios and count features retain their direct scales but are still interpreted jointly with those relative columns. SMOTE's nearest-neighbor interpolation mixes complete feature vectors from *different* events, producing synthetic samples whose relative and absolute components have no single corresponding cross-section in which they are coherent. The synthetic samples can be plausible marginally but out-of-distribution jointly with respect to the panel structure.
3) **High-dimensional sparsity.** With 81 features and only 290 positive training samples, the nearest-neighbor graph that SMOTE relies on is sparse and noisy: each minority point has few true neighbors in its class, and interpolation typically bridges across different subpopulations (e.g., small-cap vs. medium-cap pumps). This is the regime in which Blagus and Lusa [21] showed that SMOTE is at best neutral and frequently *degrades* classification performance.

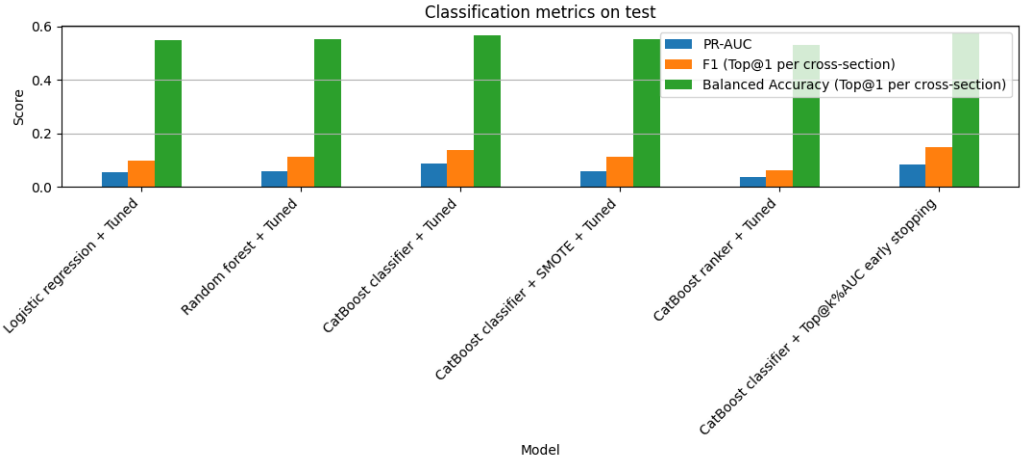


FIGURE 9. **Standard classification metrics on the test set for all tuned models and CatBoost + ES. Each grouped bar reports Precision-Recall AUC (left), F1-score under a Top@1-per-cross-section decision rule (middle), and balanced accuracy, the average of sensitivity and specificity (right). PR-AUC is preferred over ROC-AUC under the $\sim$ 1:287 class imbalance.**

By contrast, class-weight reweighting in the CatBoost loss function does not add, remove, or perturb any feature value. It only changes the per-sample gradient magnitude, concentrating the optimizer on the decision boundary between the real positive and negative samples in their native, cross-sectionally-normalized feature space. This is consistent with the observed performance of class-weight-based imbalance handling over synthetic oversampling in our setting and is consistent with the cost-sensitive-learning literature [24].

G. ANALYSIS OF RANKING MODEL PERFORMANCE

The ranker with saved tuned parameters attains test Top@k%AUC of 0.629 and Top@1 of 6.25%. It remains below CatBoost + ES (0.674) and tuned logistic regression (0.697) on aggregate AUC. Ranking advantages reported in cross-sectional equity prediction [14] therefore do not transfer automatically to this target-prediction sample.

Target-signal density. In the cross-sectional LTR setup of [14], every asset in the cross-section carries an informative continuous return target, so the listwise loss receives dense, ordinally-meaningful supervision. In our setup, exactly one asset per cross-section is a true manipulation target. All other assets contribute noise to the ordering: their 5-minute post-announcement returns are driven by background market movements, independent news, or incidental volatility, none of which the model is asked to predict. The ranker therefore spends most of its gradient on nuisance ordering among non-pumped assets, while the classifier only has to separate the single pump from the rest.

Loss-function alignment with the Top@k evaluation objective. Pairwise and listwise ranking losses (YetiRank, PairLogit, etc.) integrate pairwise or position-weighted comparisons across the entire list. When the positive cardinality is 1, most of those pairs are negative-vs.-negative and therefore non-informative for Top@k. The classification log-loss, conversely, concentrates gradient mass on the decision boundary between the single positive and the $\sim$290 negatives, which is exactly the boundary that Top@k accuracy measures. Class-weight reweighting further tilts the log-loss toward the positive class, yielding a gradient that is sample-efficient in the 1-vs.-many regime.

Target-construction noise. The continuous ranking target (realized arithmetic return over the first five minutes after announcement) is additionally noisy because (i) coincident market-wide moves can push a non-pumped asset's five-minute return above the pump target's, and (ii) micro-cap order books produce large relative returns on very small trades, which the ranker has no way to discount. In the classification formulation the binary label is immune to this type of noise.

Taken together, these three factors may help explain why, in the extreme-imbalance cross-sectional setting that characterizes P&D target prediction, class-weighted classification outperforms the evaluated ranker even though the evaluation metric is itself a ranking metric.

Among the original candidates, tuned logistic regression has the largest test Top@k%AUC point estimate (0.697), followed by CatBoost + ES (0.674), tuned CatBoost (0.662), and tuned random forest (0.651). Section IV-I adds a comparison of tuned models with shared resamples and multiplicity correction; CatBoost + ES remains the validation-selected model.

H. INTERPRETATION OF THE SELECTED MODEL

We use CatBoost PredictionValuesChange to describe the selected model's fitted feature importance. Fig. 10 shows the full-training fit. Prior-pump history and multi-day trade activity are prominent, alongside return and order-flow features. These importance scores describe the fitted model and are not causal effects or evidence of manipulation by themselves.

The same importances are recorded in each of the 25 random 70% training-subset fits of Section IV-J. Table VIII reports the 15 highest median importances. Prior-pump his-

tory leads, followed by 14-day, 7-day, and 2-day trade counts and 4-hour returns. The broad history/activity pattern persists, although feature ordering varies across fits.

I. STATISTICAL SIGNIFICANCE

1) CatBoost paired comparison

Table IX retains the original intervals for CatBoost + ES and the validation-selected tuned CatBoost comparator. Their observed AUC difference is 0.0116, with paired 95% CI $[-0.0120, 0.0369]$ and one-sided $p = 0.1739$ from 2000 paired event resamples. The original comparison does not establish an aggregate-ranking advantage at the 5% level.

2) Event resampling of tuned models

For the tuned models in the bootstrap, validation Top@k%AUC is 0.646 for CatBoost + ES, 0.611 for L2 logistic regression, and 0.610 for tuned L1. To quantify test-sample uncertainty, we draw 10,000 samples of 80 whole pump events with replacement (seed 42), using identical draws for all seven fixed fitted tuned models. Repeated events retain their multiplicity. Top@k%AUC is recomputed from each resampled event set; models are not retrained within this bootstrap. Resampling candidate rows separately would ignore their within-event dependence. The event bootstrap assumes independent events and does not account for remaining temporal dependence, training uncertainty, or earlier inspection of the test period.

Table X reports all model point estimates and pointwise 95% percentile intervals. The normalised Top@k%AUC integrates the curve from 0 to 20% in 0.5% increments. Pairwise differences use a two-sided bootstrap null distribution centred at zero, an add-one Monte Carlo correction, and Holm adjustment across all 21 unordered model pairs. Marginal-interval overlap is not used as a significance test.

The L1-minus-L2 test Top@k%AUC difference is -0.0080, with paired 95% CI $[-0.0264, 0.0113]$ and Holm-adjusted $p = 1.000$. These comparisons do not establish superiority or equivalence; the primary model is chosen by validation Top@k%AUC.

CatBoost + ES minus tuned L1 logistic regression has observed Top@k%AUC difference -0.0048, paired 95% CI $[-0.0511, 0.0388]$, and two-sided $p = 0.8368$ (Holm-adjusted $p = 1.000$). Against L2 logistic regression, the difference is -0.0128, CI $[-0.0630, 0.0345]$, two-sided $p = 0.6063$ and adjusted $p = 1.000$. Neither comparison establishes superiority or equivalence. CatBoost + ES is significantly above tuned random forest, tuned SMOTE, and the tuned ranker after adjustment, but not above every candidate. Fig. 12 shows the tuned-model AUC intervals; the complete AUC distributions, paired comparisons, predictions, and event draws accompany the reproduction bundle.

J. ROBUSTNESS ANALYSIS

1) Training data perturbation

Twenty-five fits of the L2 logistic comparator on random 70% training-event subsets give mean test Top@k%AUC 0.707, standard deviation 0.007, and a central 80% interval $[0.698, 0.715]$. The selected L2 penalty, C, class weights and validation/test membership remain fixed. These are conditional training-sensitivity results, not a new model-selection experiment; the full distributions accompany the reproduction bundle.

We assess the stability of the selected CatBoost + ES model by retraining it 25 times, each time using a random 70% subset of the training cross-sections while keeping the validation and test sets fixed. Table XI summarizes the distribution of test-set metrics across runs.

The Top@k%AUC standard deviation is 0.013 across the 25 subset fits, with a central 80% interval of $[0.662, 0.693]$. Fig. 13 visualises this conditional sensitivity to training composition. These subset fits are not independent external validation and do not establish future-regime stability.

2) Temporal subperiod analysis

Our test set spans from May 2021 to August 2022, covering the late-2021 bull market and the onset of the 2022 contagion around the Terra/LUNA collapse. We conducted the subperiod analysis with June 1, 2022 chosen *ex ante* as the bull→bear regime boundary, so that the split is not a function of the model's test-set performance.

The early subperiod [May 2021, June 1, 2022) contains 76 pump events; CatBoost + ES attains Top@k%AUC 0.676. The late subperiod contains only 4 events and is excluded by the fixed `min_pumps=10` rule. Consequently this sample does not support a reliable quantitative claim about sustained bear-market performance or cross-regime generalisation.

V. PORTFOLIO STRATEGY

This section evaluates a portfolio strategy inspired by [16], with modified construction and execution assumptions tailored to the proposed scoring framework. Authors of [16] proposed using trained classification models to produce logits for each asset in the cross-section, then construct a portfolio of assets that have their logit of being pumped above some threshold. They also proposed using weights of the portfolio proportional to the logits produced by the model. They backtested their strategy using the following assumptions: assets that are not pumped do not change in price over the pump, authors buy at open price of the last hourly candle prior to the pump and are able to sell at half of pump's max return. They backtested their strategy using the test sample from October 29, 2018 to January 11, 2019 and were able to achieve a return of 60% over approximately two months. We extend this approach by constructing portfolios from the proposed model scores and by adopting more conservative execution assumptions.

We propose using models tuned for Top@k%AUC which will ensure that the models are optimized not only for Top@1 accuracy but for any arbitrary values of k. We relaxed some assumptions of [16] in the way we trade our portfolios, the timeline of how we trade each pump is described below:

1) Rank tickers by model logits (descending).

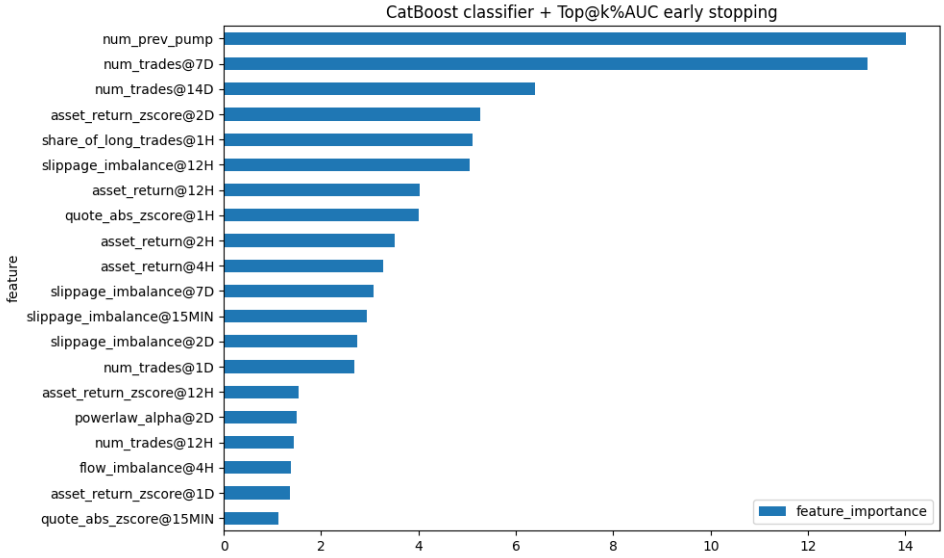


FIGURE 10. **Feature importances for the CatBoost classifier + Top@k%AUC early stopping. Bars report CatBoost PredictionValuesChange for the full-training fit; the leading features are prior pump history, 7-day and 14-day trade counts, two-day standardized return, and the one-hour share of long trades.**

TABLE VIII. **Training-subset distribution of CatBoost + ES feature importances (PredictionValuesChange) across the 25 random 70% training-subset retraining runs of Section IV-J. The 15 features with the highest median importance are shown; columns report per-feature percentiles and rows are sorted by median.**

Feature	10%	25%	Median	75%	90%
num_prev_pump	8.74	12.58	16.28	20.57	25.88
num_trades@14D	2.69	4.03	7.11	8.68	13.04
num_trades@7D	2.55	3.72	5.92	8.67	10.91
num_trades@2D	0.50	2.17	5.02	8.57	16.60
asset_return@4H	0.45	1.84	2.68	4.12	4.82
slippage_imbalance@12H	0.00	0.83	2.60	4.87	8.76
slippage_imbalance@2D	0.44	1.12	2.35	3.83	7.52
asset_return@2H	0.66	1.80	2.25	4.08	5.37
quote_abs_zscore@1H	0.26	1.14	2.24	2.95	4.49
asset_return@12H	0.19	1.22	2.06	2.62	4.65
num_trades@1D	0.00	0.61	1.83	2.92	4.02
asset_return_zscore@12H	0.00	0.48	1.73	2.63	4.36
asset_return_zscore@1H	0.06	0.78	1.69	2.70	4.09
asset_return_zscore@4H	0.03	0.57	1.51	2.85	3.53
asset_return@1D	0.00	0.66	1.48	1.95	3.04

TABLE IX. **Top@k%AUC on the test set with 95% cross-section-level bootstrap confidence intervals (1000 iterations). Top@k%AUC is integrated and normalized over $k\% \in (0, 20\%]$ (Eq. 6).**

Model	Top@k%AUC	95% CI
CatBoost + ES	**0.674**	[**0**.592, **0**.755]
CatBoost classifier + Tuned	0.662	[0.575, 0.747]

2) Select the Top@k set of tickers.
3) Submit equal-weight entries 15 minutes before the announcement and fill each at its first subsequent pre-announcement trade; omit a leg if no such trade occurs. The omitted leg's fixed $1/k$ weight remains in cash and is not redistributed.
4) Hold until the announcement; if not pumped, sell at the announcement; if pumped, sell 1 minute after the pump.
5) Liquidate the complete acquired base-asset quantity, convert the BTC proceeds to USDT at the exit-time completed indicative rate, and subtract a 25 bp (0.25%) round-trip cost from entry capital.

We use CatBoost + ES as the portfolio model selected by validation Top@k%AUC. The execution-aware sensitivity analysis retains the original $k = 5$ specification; test results for $k \in \{1, 2, 5, 10, 20, 30\}$ are descriptive concentration

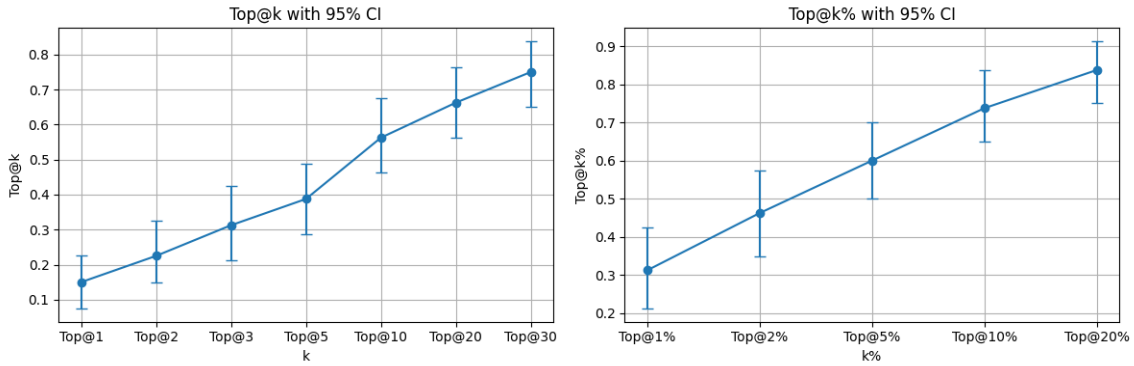


FIGURE 11. **Cross-section-level bootstrap 95% confidence intervals for CatBoost + ES on the test set, based on 1000 resamples of the 80 test cross-sections with replacement. Left: Top@k accuracy with error bars at $k \in \{1, 2, 3, 5, 10, 20, 30\}$. Right: Top@k% accuracy with error bars at $k\% \in \{1, 2, 5, 10, 20\}$. Takeaway: the point estimates lie above random, while the interval widths argue for cautious interpretation of narrow differences between models.**

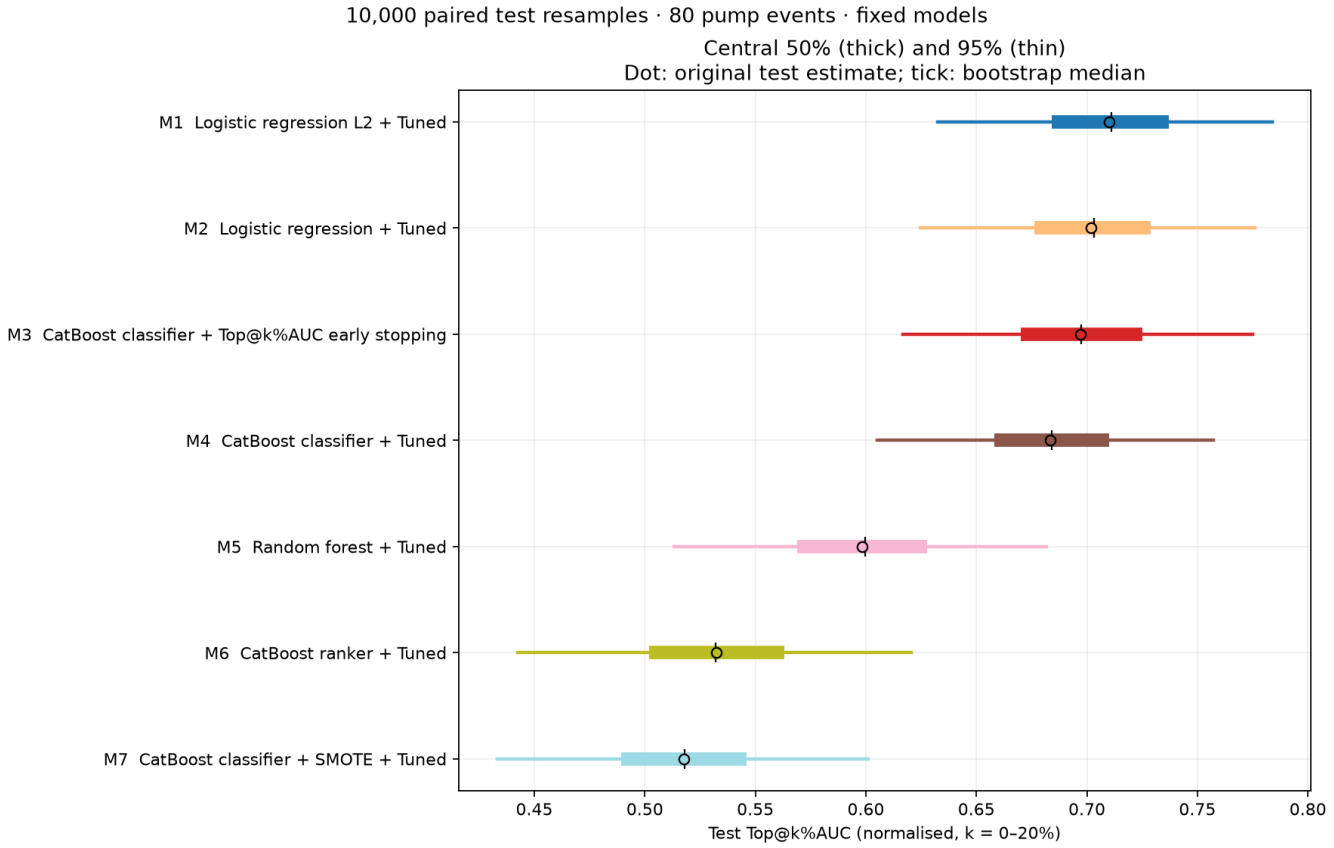


FIGURE 12. **Test Top@k%AUC intervals from 10,000 paired resamples of the same 80 pump events for the seven fixed tuned models: central 50% (thick) and 95% (thin) bootstrap intervals, unresampled test estimates (dots), and medians (ticks). Intervals are pointwise; significance uses paired differences and Holm correction across 21 comparisons, not interval overlap.**

checks and are not used to retune portfolio settings. The test period was inspected in earlier research, so these simulation results remain exploratory.

To benchmark asset selection, we generate 10,000 random portfolios at each k (seed 42). At every test event, k distinct eligible assets are selected uniformly without replacement and assigned weight $1/k$, using common random asset orders across portfolio sizes. Entry/exit rules, USDT conversion and the 25 bp round-trip cost match the CatBoost strategy; unfilled allocations remain in cash. The 80 events stay fixed in chronological order, and this comparison excludes size-dependent impact. Returns are accumulated without reinvestment. Annualized statistics use zero-filled daily returns over the same calendar and 365 days per year; return, volatility

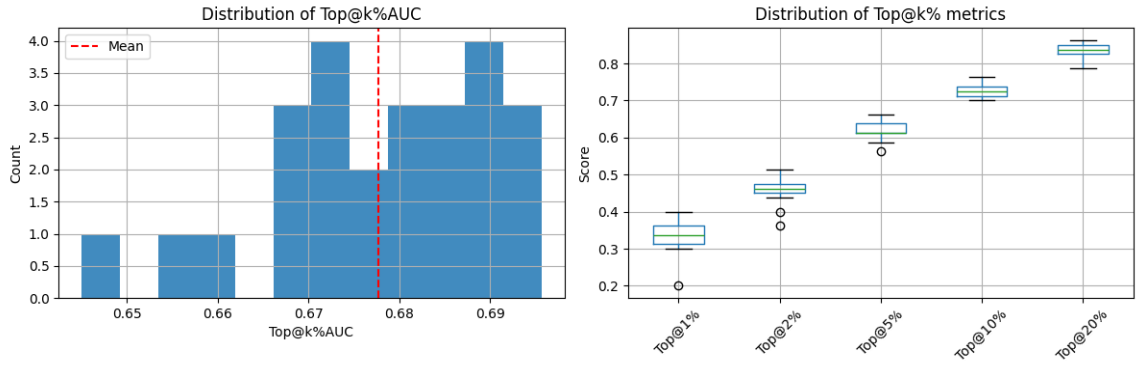


FIGURE 13. **Distribution of test-set metrics of CatBoost + ES across 25 independent retraining runs. Each run uses a different random 70% subset of the training cross-sections; the validation and test sets are held fixed. Left: histogram of Top@k%AUC (integrated and normalized over $k\% \in (0, 20\%]$) across the 25 runs; the vertical dashed line is the across-run mean. Right: box plots of Top@k% at $k\% \in \{1, 2, 5, 10, 20\}$. Takeaway: $\sigma_{\text{Top@k\%AUC}} = 0.013$ and the central 80% of runs fall in $[0.662, 0.693]$.**

TABLE X. **Top@k%AUC on the test set with 95% cross-section-level bootstrap confidence intervals (10,000 paired resamples). Top@k%AUC is integrated and normalized over $k\% \in (0, 20\%]$ (Eq. 6).**

Model	Top@k%AUC	95% CI
Logistic regression L2 + Tuned	0.710	[0.633, 0.784]
Logistic regression + Tuned	0.702	[0.625, 0.776]
CatBoost + ES	0.697	[0.617, 0.775]
CatBoost + Tuned	0.683	[0.605, 0.757]
Random forest + Tuned	0.598	[0.513, 0.682]
CatBoost ranker + Tuned	0.532	[0.442, 0.621]
CatBoost + SMOTE + Tuned	0.518	[0.433, 0.601]

TABLE XI. **Robustness analysis: distribution of test-set metrics across 25 runs with random 70% training subsets. Top@k%AUC is integrated and normalized over $k\% \in (0, 20\%]$ (Eq. 6). The standard deviation of Top@k%AUC is 0.013.**

Metric	Mean	Std	[10%, 90%]
Top@k%AUC	0.678	0.013	[0.662, 0.693]
Top@1%	0.334	0.038	[0.305, 0.370]
Top@2%	0.463	0.032	[0.438, 0.500]
Top@5%	0.619	0.023	[0.593, 0.645]
Top@10%	0.727	0.016	[0.705, 0.745]
Top@20%	0.839	0.017	[0.825, 0.858]

and Sharpe are computed for each draw before averaging. The plotted mean path is an average across selections, not a single traded portfolio.

Table XII and Fig. 14 show that the descriptive Sharpe ratio peaks at the retained $k = 5$ specification (2.66). The $k = 2$ portfolio has the largest no-reinvestment cumulative return (1.453). The six simulated CatBoost portfolios have cumulative returns ranging from 0.233 to 1.453, compared with a 57.3% BTC/USDT decline over the same calendar. This concentration comparison does not change the $k = 5$ impact-sensitivity specification or establish executable profits; returns use fixed event notional and stated fill assumptions.

For $k = 5$, the random reference has mean cumulative return -10.22% and a central 95% selection range of $[-23.29\%, 20.90\%]$, compared with 107.89% for CatBoost + ES. Its mean Sharpe is -1.46 (95% range $[-3.22, 0.62]$), versus 2.66 for CatBoost; mean target inclusion is 1.74%, versus 45.00%. These ranges describe random asset selection conditional on the observed test period, not event-bootstrap confidence intervals or confirmatory significance tests; independent holdout assessment remains necessary.

For market context, Fig. 14 includes a continuous BTC/USDT buy-and-hold position over the exact strategy calendar, from May 1, 2021 16:45:08 UTC to August 14, 2022 16:01:05 UTC. It enters at the first decision timestamp ($T-15$ minutes) and is valued finally at the last strategy exit ($T+1$ minute), using completed one-minute BTC/USDT VWAPs. Daily UTC observations connect those exact endpoints. Each plotted price increment is expressed relative to the fixed initial BTC price, so its cumulative sum equals the simple full-period BTC return exactly. BTC/USDT records cumulative return -0.573, annualized return -0.445, daily annualized volatility 0.734, and Sharpe ratio -0.606. This is a calendar-aligned market benchmark rather than an exposure-matched strategy: BTC remains invested between events, whereas each Top@k portfolio is exposed only during its event window.

Portfolio / benchmark	Avg. event return	Cumulative return	Annualized portfolio return	Annualized portfolio volatility	Sharpe ratio
CatBoost + ES $k = 1$	0.0097	0.7747	0.6004	0.7319	0.8203
CatBoost + ES $k = 2$	0.0182	1.4529	1.1259	0.5143	2.1892
CatBoost + ES $k = 5$	0.0135	1.0789	0.8361	0.3141	2.6622
CatBoost + ES $k = 10$	0.0079	0.6311	0.4890	0.1926	2.5396
CatBoost + ES $k = 20$	0.0043	0.3441	0.2667	0.1271	2.0976
CatBoost + ES $k = 30$	0.0029	0.2332	0.1807	0.0888	2.0349
Random $k = 1$	-0.0013	-0.1023	-0.0793	0.1313	-1.2805
Random $k = 2$	-0.0013	-0.1015	-0.0787	0.1089	-1.3999
Random $k = 5$	-0.0013	-0.1022	-0.0792	0.0843	-1.4597
Random $k = 10$	-0.0013	-0.1029	-0.0798	0.0699	-1.4881
Random $k = 20$	-0.0013	-0.1033	-0.0801	0.0587	-1.5473
Random $k = 30$	-0.0013	-0.1035	-0.0802	0.0536	-1.6031
BTCUSDT buy-and-hold	—	-0.5727	-0.4448	0.7340	-0.6060

TABLE XII. Validation-selected CatBoost + ES portfolio performance by size without size-dependent price impact, and the time-aligned BTC/USDT benchmark. Random rows are means of the statistics computed separately for 10,000 draws; all returns are decimal fractions.

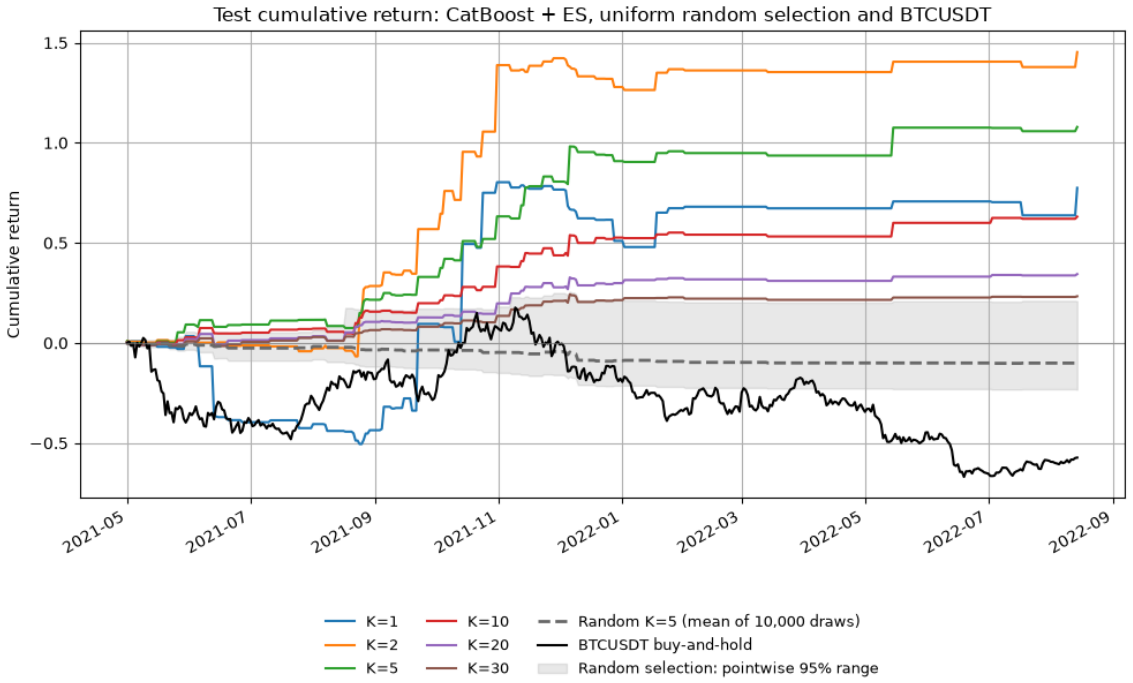


FIGURE 14. Cumulative return paths on the test set for portfolios of size $k \in \{1, 2, 5, 10, 20, 30\}$, ranked by validation-selected CatBoost + ES scores, together with a continuous BTC/USDT buy-and-hold benchmark over the identical calendar interval. Portfolio returns are accumulated without reinvestment and include a 25 bps round-trip cost. The BTC benchmark is the solid black line. The dashed gray line is the mean of 10,000 uniformly sampled $k = 5$ portfolios; shading is the pointwise 95% range across selections. Size-dependent price impact is excluded here.

A. EXECUTION-AWARE BACKTESTING WITH PRICE IMPACT

The results above use the first observed trade at or after each execution threshold but do not otherwise charge size-dependent slippage, which is unrealistic for the illiquid small-cap tokens typically targeted by P&D operations. To assess practical feasibility, we extend the backtesting framework with a fitted market-impact model that produces *size-dependent, dynamic* slippage in addition to the fixed round-trip fee. The model serves two purposes: first, it estimates a volume-weighted average price (VWAP) for both entry and exit; second, the shape of the impact curve is itself the *liquidity constraint* of the strategy—larger orders remain executable, but the fitted $\beta\sqrt{Q}$ curve makes each additional unit of notional arrive at progressively worse prices, so the economically relevant question shifts from "is the order ex-

ecutable?" to "at what order size does impact erode the signal?".

1) Impact model specification

We model price impact in basis points as a function of order notional Q (in USDT):

$$I(Q) = \beta\sqrt{Q}. \tag{8}$$

The square-root specification is a well-established empirical regularity in market microstructure. Kyle [25] introduced the foundational model of linear price impact from informed order flow; subsequent work demonstrated that the aggregate impact of executed orders follows a concave, approximately square-root, relationship with order size. Almgren and Chriss [19] formalized this within the optimal execution framework, while Almgren et al. [20] provided direct empirical estimates of the power-law exponent (≈ 0.6) using institutional equity data. Bouchaud et al. [22] surveyed the evidence for the square-root law across asset classes, and Tóth et al. [28] demonstrated its universality across stocks and time periods using a large meta-order dataset. In cryptocurrency markets specifically, Donier and Bonart [23] confirmed the square-root law in Bitcoin using over one million reconstructed meta-orders, showing that the relationship holds across four decades of order sizes.

We fit a single coefficient $\beta \geq 0$ via constrained OLS (no intercept) for each asset, pooling absolute impacts from both buy and sell candles. The entry regression uses the 14 days of 5-minute candles ending strictly at the actual simulated entry; it therefore cannot observe any later trade. For the manipulated asset, the execution backtest fits 5-second sell-dominated candles from T up to, but excluding, the realized exit timestamp, capped at $T + 10$ minutes. In each case, absolute net volume serves as Q and the absolute close-to-close price change (in bps) serves as the impact observation. Entry notionals are converted to USDT using the most recent completed BTC/USDT one-minute bar at entry; the exit model uses the completed rate available at the announcement. Neither conversion falls forward to a future or incomplete bar.

2) Impact measurement from candle data

Entry model (5-minute candles). We aggregate the 14-day trade history ending strictly at the actual simulated entry timestamp into 5-minute candles. For each candle we compute the net buy volume $Q_{\text{net}} = 2 \times V_{\text{taker buy}} - V_{\text{total}}$ (in quote currency), where $V_{\text{taker buy}}$ is the taker-buy quote volume and V_{total} is the total quote volume. The absolute net volume $|Q_{\text{net}}|$ serves as the order size, and the absolute close-to-close price change $|p_{\text{close}}/p_{\text{prev close}} - 1| \times 10^4$ bps serves as the impact observation. Observations from both buy- and sell-dominated candles are pooled into a single regression, which doubles the effective sample size and produces a more stable estimate of β. The 5-minute resolution matches the intended execution horizon: we assume the order is worked over a single candle interval, so the candle-level impact directly estimates the cost of execution.

Exit model for manipulated assets (5-second candles, sell-only). For the pumped asset, the backtest fits a separate model using only sell-dominated candles (negative Q_{net}) in $[T, \min(T + 10\text{min}, t_{\text{exit}}))$. The right-open exit bound prevents post-exit observations from entering the simulated execution cost. The full 10-minute window is used only for the illustrative regression figure below.

We model liquidity through a continuous price-impact penalty rather than a hard participation cap. This makes larger orders executable only at increasingly adverse VWAPs. A hard order-book depth or participation-rate cap would further reduce realized returns and is therefore a conservative extension for future work.

3) VWAP entry and exit price estimation

The execution simulation proceeds as follows for each selected asset:

Step 1: Determine raw prices. Once the model has scored the cross-section at $t_{\text{pump}} - \Delta t_{\text{buy}}$, the entry price p^{entry} is the first trade price at or after that decision timestamp and strictly before t_{pump}. If no such trade occurs, the leg is not executable and is omitted. The exit price p^{exit} is the first trade price at or after t_{pump} (or $t_{\text{pump}} + \Delta t_{\text{sell}}$ for the manipulated asset). Thus no simulated fill predates the signal that generated it.

Step 2: Resolve intended notional. The target order size Q_{intended} is specified in USDT and converted to quote currency (BTC) using the indicative BTC/USDT rate at entry time. This normalization ensures that intended position sizes are comparable across BTC price regimes.

Step 3: Track inventory and exit notional. Let $R^{\text{entry}}_{\text{BTC/USDT}}$ denote the completed indicative conversion rate at entry. The entry quote notional and acquired base-asset quantity are

$$Q^{\text{entry}}_{\text{BTC}} = Q_{\text{intended}}/R^{\text{entry}}_{\text{BTC/USDT}}, \qquad B = Q^{\text{entry}}_{\text{BTC}}/p^{\text{entry}}_{\text{vwap}}. \tag{9}$$

The exit liquidates all B units. Its impact is evaluated at the contemporaneous raw quote notional $Q^{\text{exit}}_{\text{BTC}} = Bp^{\text{exit}}$, rather than reusing the entry BTC notional. Internally the impact functions use USDT-normalized sizes $\widetilde{Q}^{\text{entry}} = Q^{\text{entry}}_{\text{BTC}} R^{\text{entry}}_{\text{BTC/USDT}}$ and $\widetilde{Q}^{\text{exit}} = Q^{\text{exit}}_{\text{BTC}} R^{T}_{\text{BTC/USDT}}$, where R^T is the completed announcement-time rate used to fit the manipulated exit model. This prevents exit slippage from being understated when the target price rises. We assume that deeper layers of the order book remain available beyond the volume observed in historical candles. Larger trades therefore remain feasible, but they incur larger slippage through the impact model.

Step 4: Compute VWAP execution prices. The fitted coefficient estimates the price displacement associated with a given net order flow. An order that walks through the book continuously fills at improving prices along the way.

Integrating the square-root impact profile over the execution path yields the average fill cost:

$$I_{\text{vwap}}(Q) = \frac{1}{Q}\int_0^Q \beta\sqrt{v}\,dv = \tfrac{2}{3}\,\beta\sqrt{Q}. \tag{10}$$

The VWAP impact is two-thirds of the terminal impact because early fills occur at better prices than the final fill level. The VWAP execution prices are then:

$$p_{\text{vwap}}^{\text{entry}} = p^{\text{entry}}\left(1 + I_{\text{vwap}}(\widetilde{Q}^{\text{entry}})/10^4\right), \tag{11}$$

$$p_{\text{vwap}}^{\text{exit}} = p^{\text{exit}}\left(1 - I_{\text{vwap}}(\widetilde{Q}^{\text{exit}})/10^4\right). \tag{12}$$

Buying pushes the fill price above the initial quote; selling pushes it below. Let $R_{\text{BTC/USDT}}^{\text{exit}}$ be the completed exit-time conversion rate. The fully USDT-denominated transaction return is

$$r_{\text{USDT}} = \frac{B p_{\text{vwap}}^{\text{exit}} R_{\text{BTC/USDT}}^{\text{exit}}}{Q_{\text{intended}}} - 1 - 0.0025, \tag{13}$$

where the 25 bps term is charged against entry capital and covers the round-trip exchange fee. This accounting includes the BTC/USDT move over the holding interval and never mixes a BTC-denominated asset return with an exit-time USDT conversion.

Manipulated-asset exit. The sell leg occurs during the manipulation window where buying and selling pressure are not balanced. Its impact model uses 5-second sell-dominated candles available strictly before the exit, never observations after the simulated transaction. A manipulation-window fit is used only when at least 20 sell-dominated candle samples are available; otherwise the framework falls back to the pre-pump entry model. The ten-minute diagnostic fit below is illustrative; each execution fit uses only observations before its own exit.

4) Regression fit quality

Fig. 15 shows an example of the fitted entry impact regression for one pump asset. The absolute-impact approach pools buy and sell 5-minute candles into a single regression ($I = \beta\sqrt{Q}$, no intercept), yielding $R^2 = 0.310$ on the buy side, $R^2 = 0.355$ on the sell side, and $\beta = 0.7224$ in the example shown.

Fig. 16 is a descriptive fit over the full capped 10-minute manipulation window for one asset. This longer diagnostic window is used only to visualize the sell-side relation; the execution backtest instead truncates each fit at that transaction's realized exit. The sell-only 5-second candle approach yields $R^2 = 0.274$ with $\beta = 1.3188$ ($n = 83$ candles) in the example shown. These illustrative coefficients do not establish a general causal difference between entry and exit liquidity.

5) Unmodeled execution frictions

The fitted impact model captures average slippage against historical candle-level liquidity but intentionally omits several real-world frictions in order to keep the estimate interpretable. Specifically, we do not model (i) order-book depth exhaustion at the deepest levels (which would manifest as partial fills rather than a smooth $\beta\sqrt{Q}$ curve), (ii) queue priority and time-in-force dynamics on the exchange matching engine, (iii) exchange-side API latency between signal generation and order arrival, and (iv) the risk that other market participants detect the same pump signature and front-run the entry. Each of these frictions works in the same direction and would reduce realized PnL further, so the sensitivity numbers reported below should be read as assumption-dependent scenarios, not a validated upper bound on achievable performance.

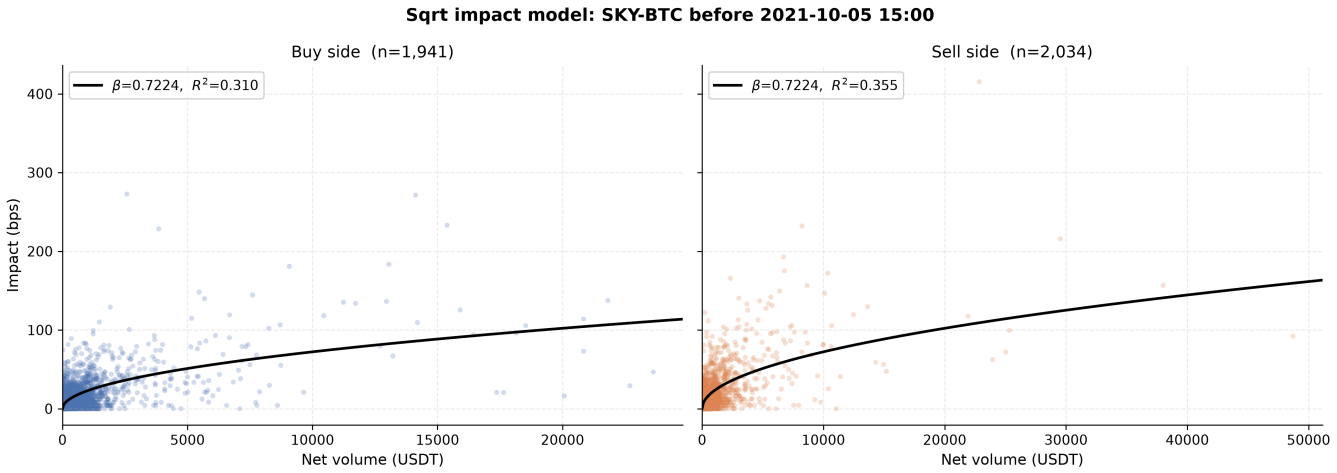


FIGURE 15. Fitted entry price-impact regression for the SKY-BTC pair, using the 14 trading days of 5-minute candles ending 15 minutes before the pump announcement. Left panel: candles with net buying pressure (positive Q_{net}); right panel: candles with net selling pressure (negative Q_{net}). Each scatter point is one 5-minute candle; horizontal axis $|Q_{\text{net}}|$ is absolute net volume in USDT and vertical axis $|\acute{p}|$ is the absolute close-to-close price change in basis points. Solid line: fitted $|\acute{p}| = \beta\sqrt{|Q_{\text{net}}|}$ (no intercept, $\beta = 0.72$), estimated on both panels jointly. Takeaway: the single-parameter square-root fit captures the concave impact–size relation on both sides of the book, consistent with the square-root impact law documented across asset classes [23], [28]; this is the curve used to translate order size into VWAP slippage in the backtest.

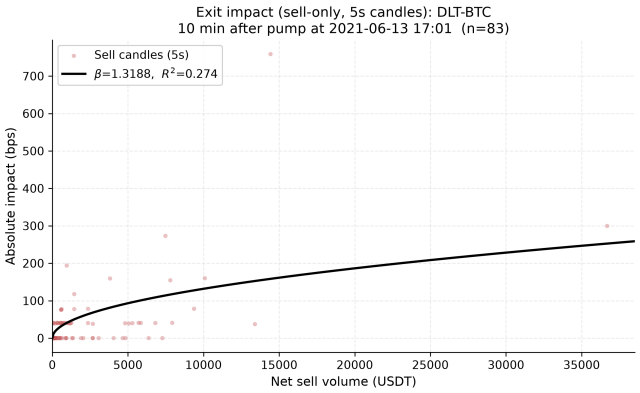


FIGURE 16. Descriptive exit price-impact regression for the DLT-BTC pair, using 5-second candles from the first 10 minutes after the pump announcement and restricted to sell-dominated candles ($n = 83$). This full-window illustration is not the execution fit: each backtested exit uses only candles strictly preceding its own exit timestamp. Horizontal axis: absolute sell-side net notional $|Q_{\text{net}}|$ in USDT. Vertical axis: absolute close-to-close price change in basis points. Solid line: fitted $|\acute{p}| = \beta\sqrt{|Q_{\text{net}}|}$ ($\beta = 1.32$). Takeaway: the exit coefficient is nearly twice the entry coefficient (1.32 vs. 0.72 on the earlier example), which illustrates different fitted coefficients in these two examples rather than a general estimate of the relative cost of entry and exit.

6) Portfolio performance with price impact

We repeat the portfolio comparison in Table XII and Fig. 14 with the fitted price-impact model enabled on both entry and exit, using an intended position of 1,000 USDT per selected asset. A size-k portfolio therefore targets $k \times 1{,}000$ USDT at each event; returns are normalized by this intended capital, with unfilled allocations retained as cash. The 80 test events, model predictions, random asset selections, entry/exit rules, and 25 bp round-trip cost are identical to the preceding comparison. Thus the change in returns isolates the modeled effect of price impact at this order size.

Table XIII and Fig. 17 report the resulting metrics and cumulative paths. At $k = 5$, CatBoost + ES cumulative return decreases from 107.89% to 71.19%, and Sharpe decreases from 2.66 to 1.81. The random $k = 5$ reference has mean cumulative return -32.99% and a central 95% selection range of $[-46.11\%, -2.22\%]$, with mean Sharpe -3.38. The largest model cumulative return remains at $k = 2$ (108.51%), while the $k = 20$ and $k = 30$ portfolios fall to -0.49% and -10.34%. These descriptive comparisons retain the original portfolio settings and the same continuous BTC/USDT benchmark.

Portfolio / benchmark	Avg. event return	Cumulative return	Annualized portfolio return	Annualized portfolio volatility	Sharpe ratio
CatBoost + ES $k = 1$	0.0049	0.3937	0.3051	0.7279	0.4191
CatBoost + ES $k = 2$	0.0136	1.0851	0.8409	0.5021	1.6747
CatBoost + ES $k = 5$	0.0089	0.7119	0.5516	0.3041	1.8142
CatBoost + ES $k = 10$	0.0033	0.2662	0.2063	0.1838	1.1221
CatBoost + ES $k = 20$	-0.0001	-0.0049	-0.0038	0.1267	-0.0301
CatBoost + ES $k = 30$	-0.0013	-0.1034	-0.0801	0.0911	-0.8798
Random $k = 1$	-0.0041	-0.3300	-0.2557	0.1394	-2.8156
Random $k = 2$	-0.0041	-0.3293	-0.2552	0.1174	-3.1083
Random $k = 5$	-0.0041	-0.3299	-0.2556	0.0932	-3.3821
Random $k = 10$	-0.0041	-0.3307	-0.2562	0.0792	-3.6087
Random $k = 20$	-0.0041	-0.3311	-0.2566	0.0687	-3.8887
Random $k = 30$	-0.0041	-0.3313	-0.2567	0.0641	-4.0733
BTCUSDT buy-and-hold	—	-0.5727	-0.4448	0.7340	-0.6060

TABLE XIII. Portfolio performance with 1,000 USDT per selected asset, fitted price impact on both execution legs, and a 25 bp round-trip cost. Random rows are means of statistics computed separately for 10,000 draws; all returns are decimal fractions. The continuous BTC/USDT market reference is unchanged.

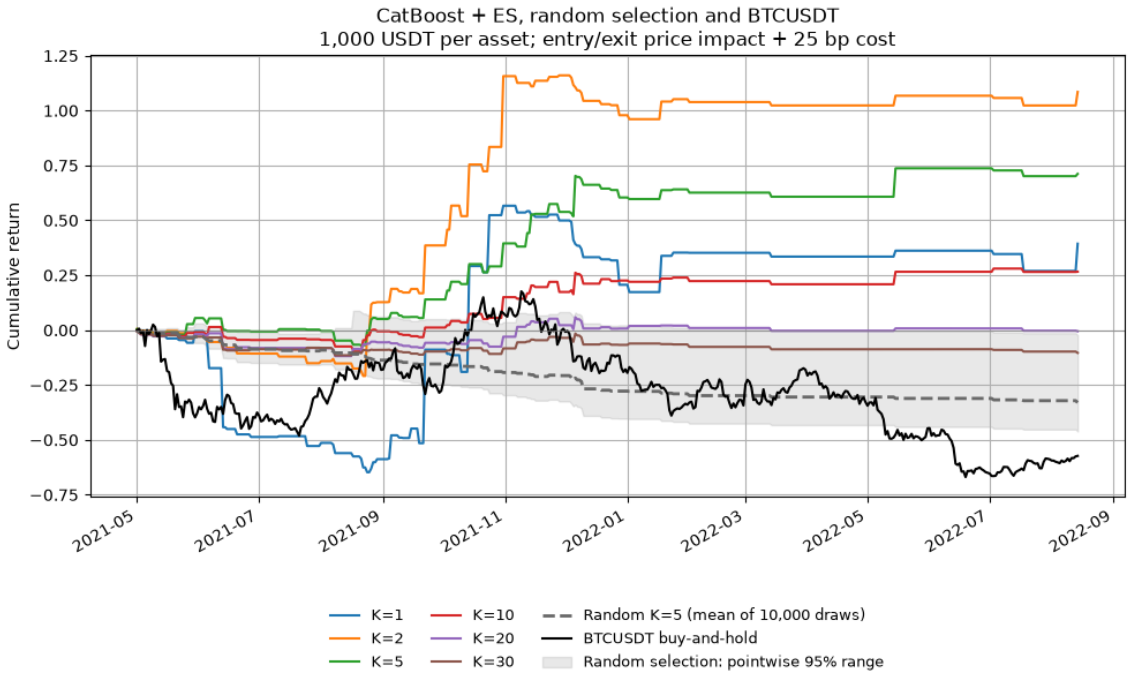


FIGURE 17. Cumulative return paths for the same portfolios and test events as Fig. 14, with fitted entry/exit price impact enabled at 1,000 USDT per asset ($k \times 1{,}000$ USDT intended portfolio capital) and a 25 bp round-trip cost. Returns are accumulated without reinvestment. The dashed gray line is the mean of the same 10,000 uniform random $k = 5$ selections, now executed with price impact; shading is their pointwise 95% selection range. The solid black line is the unchanged continuous BTC/USDT buy-and-hold reference.

7) PnL sensitivity to order size

We sweep intended order size from 100 to 10,000 USDT for the retained $k = 5$ CatBoost + ES portfolio, with a 15-minute pre-announcement decision and a 1-minute post-announcement exit. Fig. 18 reports no-reinvestment cumulative ROE over 80 events under the candle-based impact model. It declines from 0.963 at 100 USDT to −0.073 at 10,000 USDT; ROE is 0.561 at 2,000 USDT and 0.262 at 5,000 USDT. These are conditional simulation outcomes, not estimates of executable capacity. Order-book reconstruction and latency-aware paper trading would be needed to assess feasibility even at small notionals.

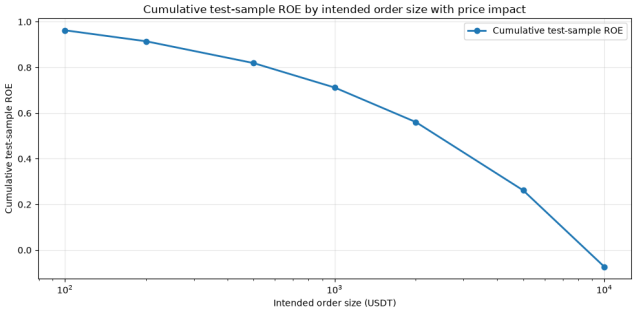


FIGURE 18. Cumulative return on invested capital (ROE) on the test sample as a function of intended per-trade order size, with the fitted candle-based square-root impact model enabled on both legs. Horizontal axis (log scale): intended order size in USDT, swept from 100 to 10,000. Vertical axis: cumulative ROE across the 80 test-set events for the $k = 5$ CatBoost + ES portfolio with a 15-minute pre-announcement decision and full inventory liquidation at the 1-minute post-announcement exit. Takeaway: cumulative ROE declines from 0.963 at 100 USDT to −0.073 at 10,000 USDT under the stated execution assumptions.

VI. LIMITATIONS AND FUTURE WORK

Exchange generalizability. Our analysis is restricted to Binance BTC pairs during 2018–2022. The evidence does not establish generalisation to USDT pairs, other exchanges, decentralised venues, later market regimes, or continuous surveillance when announcement times are unknown. Each extension requires new labels, point-in-time features, and out-of-period or external validation.

Dataset size. With 473 analyzed events (80 in test), statistical power remains limited. The reported intervals condition on the fitted models and assume independent pump events; correlated campaigns and repeated test inspection may make them optimistic. No statistically significant difference between either tuned logistic variant and CatBoost + ES is established. The validation-selected model requires new holdout validation because the test period was inspected in earlier research.

Execution assumptions. The fitted impact and 25 bp cost assumptions do not establish real trading feasibility. Order-book depth, queue priority, partial fills, signal/API latency, front-running, and exchange access restrictions are incompletely observed. Historical transaction prices do not guarantee executable quotes or scalable capacity. Predictive hit rates and simulated PnL are therefore separate outcomes, and positive simulated returns are not evidence of attainable live profits.

Evolving manipulation tactics. Manipulators may adapt their strategies over time. The late test subperiod has too few events for a reliable comparison, so our subperiod analysis does not establish cross-regime stability. Later chronological holdouts and monitoring would be needed for deployment.

Alternative imbalance-handling approaches. While we compared class weighting and SMOTE, other techniques such as focal loss [26], cost-sensitive learning [24], and deep anomaly detection [27] remain unexplored in this setting and could potentially improve performance.

Temporal safeguards and reproducibility. The pipeline enforces explicit temporal information boundaries to prevent look-ahead: predictor windows are decision-anchored and half-open, ending at $T - 15$ minutes; normaliser history is strictly pre-cutoff; and imputation priors are fitted on training events only. Automated regression tests inject extreme future trades, future-only listings and later pump announcements, then verify that the eligible asset universe and all predictors remain unchanged. Additional tests check exact timestamp boundaries, training-only imputation, exclusion of outcome columns from model inputs, isolation of automated fitting and tuning from test data, completed-bar currency conversion, and exclusion of post-execution trades from impact-model fits. These checks validate the absence of future-data influence in the tested computational paths.

The resampling analyses use frozen feature files and saved hyperparameters; the candidates are refitted and their predictions and resamples are retained. The additional L2 comparator uses the validation-only search specified above. The reproduction bundle includes the 473-event cohort, its 498 source records and 25 data-availability exclusions, partition membership, frozen feature inputs, model parameters, package versions, executed notebooks, test results, metric tables, predictions, shared bootstrap draws, and the random-portfolio asset outcomes, exact selections and per-draw statistics. These records support computational auditability but cannot independently validate Telegram annotations whose messages and URLs are unavailable; the manifest and reviewed inclusion/exclusion records do not replace those primary sources.

VII. CONCLUSION

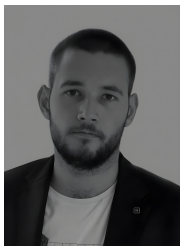
This study provides an empirical comparison of established imbalance-aware methods for event-level P&D target prediction on Binance BTC pairs. CatBoost with Top@k%AUC early stopping is selected by the highest validation Top@k%AUC and is used for portfolio analysis. L2 logistic regression has the highest test AUC point estimate in the expanded comparison, but its paired difference from CatBoost + ES is not significant; descriptive test rankings do not change the validation-based selection. The execution analysis illustrates sensitivity to assumptions and order size, not practical trading profitability.

The evaluated SMOTE and ranking variants are weaker than the leading classifiers in this sample. These results motivate further work on market surveillance, with independent label verification and external chronological validation needed before broader claims or operational use.

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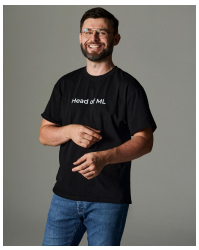
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